

Lesson 5: HTML Basics

Intro to Programming — Code the Dream

Session Overview

By the end of this session, you will be able to:

- Explain what happens when you load a web page
- Describe the roles of HTML, CSS, and JavaScript
- Identify elements, tags, and attributes in HTML code
- Write an HTML boilerplate from scratch
- Use headings, paragraphs, lists, links, and sections

Explore

How the Web Works

When you visit a website, three things happen:

1. Your **browser** sends a request to a server
2. The **server** sends back files
3. Your **browser** reads the files and shows the page

The browser is the client. It asks. The server answers.

HTML, CSS, and JavaScript

Every web page is built with three languages:

Language	Job
HTML	Structure — what's on the page
CSS	Style — how it looks
JavaScript	Behavior — what it does

This week: **HTML only**.

What Is HTML?

HTML stands for HyperText Markup Language.

- Created in 1989
- Uses **tags** inside angle brackets `< >` to describe content
- Tells the browser: "This is a heading." "This is a link." "This is a list."

```
<p>This is a paragraph.</p>
```

- `<p>` — opening tag
- `</p>` — closing tag (has a `/`)
- The text in between — the content

Elements vs. Tags vs. Attributes

Element — the whole package: opening tag + content + closing tag

```
<h1>Maria Santos</h1>
```

Self-closing tag — some elements need only one tag

```

```

Attribute — extra information inside the opening tag

```
<a href="https://github.com">GitHub</a>
```

Headings and Paragraphs

HTML has **six heading levels**. Level 1 is the biggest.

```
<h1>My Name</h1>  
<h2>About</h2>  
<h3>Subsection</h3>
```

Paragraphs use `<p>`:

```
<p>I am learning to code with Code the Dream.</p>
```

Use headings to organize your page, not just to make text big.

Lists

Unordered list — bullet points

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Ordered list — numbered

```
<ol>
  <li>Read the lesson</li>
  <li>Write the code</li>
</ol>
```

Links and Images

Link — uses `<a>` and the `href` attribute

```
<a href="https://github.com/yourusername">GitHub</a>
```

Image — self-closing, uses `src` and `alt`

```

```

Always include `alt` text. Screen readers use it.

Check for Understanding

What is the correct way to write a closing tag for a paragraph?

- A) `<p/>`
- B) `<\p>`
- C) `</p>`
- D) `<!--p-->`

The HTML Boilerplate

Every HTML file starts with this structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    <!-- visible content goes here -->
  </body>
</html>
```

- `<!DOCTYPE html>` — tells the browser this is HTML5
- `<head>` — metadata (not visible to users)
- `<body>` — everything users see

Inside the `<head>`

The `<head>` holds information *about* the page — not visible content.

```
<head>
  <title>Maria's Portfolio</title>
  <meta charset="utf-8">
  <meta name="description" content="My portfolio page">
</head>
```

- `<title>` — shows in the browser tab
- `<meta charset>` — sets the character encoding
- `<meta name="description">` — helps with search engines

Organizing with `<section>` and `id`

Use `<section>` to group related content.

Use `id` to label each section.

```
<section id="about">
  <h2>About</h2>
  <p>I'm learning to code.</p>
</section>
```

The `id` attribute lets you jump to a section or style it later with CSS.

Check for Understanding

Where does visible page content go in an HTML file?

- A) Inside `<head>`
- B) Inside `<meta>`
- C) Inside `<body>`
- D) Inside `<!DOCTYPE html>`

Check for Understanding

Which tag creates a clickable link?

```
<a href="https://github.com/yourusername">My GitHub</a>
```

- A) <link>
- B) <url>
- C) <a>
- D) <href>

Apply

Core Activity: Build a Portfolio Page

We'll build `index.html` together — **tell me what to type!**

Open **VS Code** and create a new file called `index.html`.

This is exactly what you'll build for the assignment.

Step 1 — Write the Boilerplate

Ask the group: What goes on line 1?

```
<!DOCTYPE html>
<html>
  <head>
    <title>Maria's Portfolio</title>
    <meta charset="utf-8">
  </head>
  <body>

  </body>
</html>
```

Save the file. Open it in a browser. What do you see?

Step 2 — Add a Name and About Section

Inside `<body>`, add:

```
<h1>Maria Santos</h1>

<section id="about">
  <h2>About</h2>
  <p>I'm learning to code with Code the Dream!</p>
</section>
```

Save and refresh. What changed?

Step 3 — Add an Experience Section

```
<section id="experience">
  <h2>Experience</h2>
  <ul>
    <li>Intro to Programming – Code the Dream</li>
    <li>Self-taught HTML</li>
  </ul>
</section>
```

Ask the group: What comes next?

Step 4 — Add a Connect Section

```
<section id="connect">
  <h2>Connect</h2>
  <ul>
    <li><a href="https://github.com/yourusername">GitHub</a></li>
    <li><a href="https://linkedin.com/in/you">LinkedIn</a></li>
  </ul>
</section>
```

Save and refresh. Click the links — do they work?

Extension Activity

Only if time allows and students are ready.

Extension — Add Images and Navigation

Challenge: Add a profile photo and a navigation menu.

```

```

```
<nav>  
  <a href="#about">About</a>  
  <a href="#experience">Experience</a>  
  <a href="#connect">Connect</a>  
</nav>
```

What happens when you click a nav link?

Assignment Preview

This week you'll build your own `index.html` portfolio page:

- HTML boilerplate: `<!DOCTYPE html>`, `<head>`, `<body>`
- At least two `<meta>` elements in `<head>`
- Your name in an `<h1>`
- **About** section — paragraph about you
- **Experience** section — list of skills or experiences
- **Connect** section — links to GitHub and LinkedIn
- Each section wrapped in `<section id="...">`

Push your work to a new `lesson-5` branch and open a pull request.

Tips for the Assignment

1. **Write the boilerplate first** — then add content inside `<body>`
2. **Save and refresh often** — check your page in the browser as you go
3. **Indent nested tags** — it makes your code easier to read and debug
4. **Stuck on a broken image or link?** Ask AI for hints:

"My image shows a broken icon. Here is my code: [paste]. Can you ask me 3 questions to help me find the path problem?"

Wrap-Up

Zoom chat: Rate your confidence from 1 to 5

How comfortable do you feel writing basic HTML?

Before next session:

- Install **VS Code** if you haven't already
- Work through the **lesson materials** (Odin Project or Scrimba)
- Build your **portfolio page** in `index.html`
- Push to a `lesson-5` branch and open a pull request
- Post questions in **Slack** anytime

Resources

Resource	What it's for
Odin Project — Elements & Tags	Core HTML reading
Odin Project — HTML Boilerplate	Boilerplate deep dive
Odin Project — Working with Text	Headings and paragraphs
Odin Project — Lists	ul, ol, li
Odin Project — Links and Images	a, img, attributes
Scrimba HTML & CSS Crash Course	Video walkthroughs
W3Schools HTML Head	Meta elements reference
MDN HTML Basics	Authoritative reference

Great work today!

You've written your first web page.

Questions? Reach out on Slack anytime.